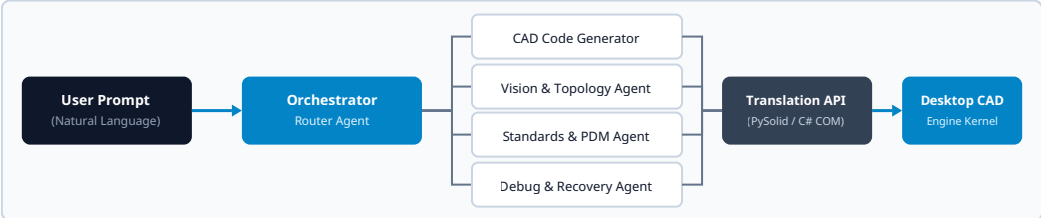


AI CAD Copilot Architecture & Ecosystem Reference

Backend Multi-Agent Execution, Plugin Mechanism, Performance Specs, and AI CAD Market Ecosystem

SECTION 1 Backend Architecture & Multi-Agent Execution System

Modern AI CAD copilots execute natural language requests through an orchestrated multi-agent framework rather than a single monolithic model:



- **Orchestrator Agent:** Parses natural language input, extracts dimensional parameters, and routes tasks to specialized agents.
- **CAD Code Generator Agent:** Translates prompts directly into native CAD API scripts (C# COM, C++, Python) rather than static mesh exports.
- **Vision & Topology Agent:** Analyzes spatial contexts, face selections, edge loops, and B-Rep structures in real-time.
- **Standards & PDM/PLM Agent:** Integrates custom enterprise title blocks, fastener hardware libraries, and DFM rulesets.
- **Debug & Recovery Agent:** Monitors rebuild status; automatically intercepts feature failures and applies code fixes.

SECTION 2 Plugin Ecosystem & Performance Specifications

Specification Domain	Technical Detail
Plugin Architecture	Lightweight desktop DLL add-in compiled in C# / .NET COM hooking directly into host CAD SDKs (SolidWorks, Inventor, Fusion 360, NX).
Execution Split	Local Host: Geometry tree extraction, active context capture, feature execution. Cloud Backend: Multi-agent LLM reasoning, code synthesis, spatial analysis via secure HTTPS/WebSockets.
Context & Token Window	Input Window: 128,000 – 200,000 tokens (accommodates large feature trees, BOMs, and design rules). Output Limits: 4,096 – 8,192 tokens per API call.
Response Latency	• Simple Operations (Filletts, Hole Patterns): 1.5 – 3.0 sec • Complex Multi-step Features: 4.0 – 8.0 sec • Full 2D Drawing Sheet Generation: 10.0 – 25.0 sec
Native File Compatibility	Direct support for fully parametric native CAD formats (<code>.SLDPRT</code> , <code>.SLDDRW</code> , <code>.IPT</code> , <code>.F3D</code> , <code>.PRT</code>).

SECTION 3 Comprehensive Ecosystem Comparison Matrix

Tool / Platform	Category	Core Capability	Ideal Technical Use Case
Leo AI	Enterprise / PDM Knowledge	3D B-Rep shape search & CAD history lookup across PLM.	Retrieving historical legacy parts, tolerance rationale, and DFM rules.
Zoo.dev (KittCAD)	Headless API Kernel	GPU-native B-Rep programmatic text-to-CAD API engine.	Automated backend parametric model creation and web CAD services.
DraftAid	Automated 2D Drafting	Learns company standard practices for auto GD&T placement.	Converting 3D models into fully dimensioned production drawings.
CoLab (AutoReview)	Design Review Agent	Automated CAD quality checks and design change graphs.	Multi-team CAD code reviews, compliance, and issue tracking.
PhysicsX / SimAI	Real-time AI Simulation	Deep learning physics inference for real-time FEA/CFD.	Instant structural and aerodynamic feedback bypassing full meshing solvers.
SolidWorks AURA	Native Desktop Copilot	In-app command execution and feature tree assistant.	Accelerating standard modeling commands inside Dassault systems.